

Automated Sand Mining Detection Using Satellite Imagery and Machine Learning

A Plain-Language Technical Documentation
of the Sand Mining Detection Pipeline

Pilot Study Area: Damodar River Basin, West Bengal / Jharkhand, India

Primary Sensor: Sentinel-2 Multispectral (10 m resolution)

Temporal Coverage: 2023–2026 (3–5 year historical window)

Mapping Scale: 0.4–0.5 km point spacing along river corridor

Ultimate Goal: All-India multi-basin sand mining probability map

What This System Does in One Sentence

It automatically looks at satellite photos of Indian rivers taken over the past 3–5 years, detects visual and statistical patterns associated with illegal sand mining, and produces a map showing exactly where mining is likely occurring right now — without anyone having to physically visit those locations.

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Introduction: The Problem This Solves

What is Riverine Sand Mining?

Sand and gravel extracted from riverbeds are the world’s most consumed solid materials by weight. In India alone, construction demand exceeds 500 million tonnes of river sand per year. The vast majority of this extraction is **illegal** — occurring without permits, environmental clearance, or any monitoring whatsoever.

Sand mining destroys riverbeds. It lowers the riverbed level (a process called “channel incision”), causes banks to collapse, cuts off groundwater recharge, kills fish and river dolphins, and can even cause bridges to collapse. Despite a legal framework (Sustainable Sand Mining Guidelines 2016, revised 2020), enforcement is almost non-existent because no one knows *where* mining is happening.

Why Can’t We Just Drive Around and Check?

India has hundreds of thousands of kilometres of rivers. The Damodar River alone — the pilot study area in this system — runs 361 km through West Bengal and Jharkhand. Physical inspection of every stretch is:

- **Impossible at scale** — too many rivers, too few inspectors
- **Dangerous** — sand mining mafias are known to be violent
- **Too slow** — illegal operations move quickly
- **Expensive** — field surveys cost lakh of rupees per campaign

What This System Does Instead

Satellites photograph every point on Earth’s surface every 5–10 days for free. This system:

1. Downloads satellite images of the river
2. Extracts mathematical features from each image (colour patterns, texture, vegetation cover, water turbidity, and how these have changed over 3–5 years)
3. Uses a machine learning model trained by a human expert to classify each location as “Sand Mining Likely”, “Possible”, or “No Mining”
4. Produces an interactive HTML map showing the results

The Satellite Data: What We See from Space

Primary Sensor: Sentinel-2

Sentinel-2 at a Glance

- **Operated by:** European Space Agency (ESA)
- **Spatial resolution:** 10 metres per pixel (visible/NIR bands)
- **Revisit time:** Every 5 days at the equator
- **Data access:** Free and open via Google Earth Engine
- **Available since:** 2017 (harmonised archive)
- **Bands used:** 10 spectral bands from Blue to SWIR

At 10 m resolution, each pixel represents a 10 m × 10 m patch on the ground. This is fine enough to see:

- Individual sand barges on a river
- Exposed sandy excavation pits
- Disturbed bank vegetation
- Turbid (muddy) water plumes from active dredging

Backup Sensors

When Sentinel-2 imagery is unavailable (e.g., heavy cloud cover), the system automatically falls back to:

Sensor	Resolution	Archive Since	Use
Sentinel-2	10 m	2017	Primary
Landsat 8/9	30 m	2013/2021	Fallback (current)
Landsat 7	30 m	1999	Historical fallback
Landsat 5	30 m	1984	Deep historical
VIIRS	375 m	2012	Turbidity / SSC proxy

Table 1: Satellite sensors used in the detection pipeline

Cloud Cover Handling

India’s monsoon season (June–September) covers most river areas with cloud 60–90% of the time. The system handles this by:

- Filtering images with < 35% cloud cover (preferred)
- Relaxing to < 60% if nothing better is available
- Using quarterly composites across 3-month windows to maximise the chance of finding a clear image in each period

Known limitation: Sentinel-1 SAR (radar) imagery that can see through clouds is not yet integrated. This is the next planned upgrade for monsoon-season detection.

Spectral Indices: Turning Pixels into Information

A satellite image records how much light is reflected from the ground in different wavelengths (colours). By combining these bands mathematically, we create “indices” — single numbers that capture specific ground conditions. Sand mining disturbs the landscape in ways that show up strongly in these indices.

The Six Core Indices

NDVI — Normalised Difference Vegetation Index

$$\text{NDVI} = \frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}$$

Measures how much green vegetation is present. Values range from -1 (no vegetation / bare soil / water) to $+1$ (dense vegetation).

Why it matters for mining: Active mining sites have vegetation stripped away. Over time, a legitimate forest or riverside field will show steadily positive NDVI. A newly mined area will show a sudden sharp drop in NDVI.

NDWI — Normalised Difference Water Index

$$\text{NDWI} = \frac{\text{Green} - \text{NIR}}{\text{Green} + \text{NIR}}$$

Detects open water bodies. High values (> 0) indicate water.

Why it matters: Helps separate in-river dredging from bankside excavation. Also helps distinguish natural sandbars from active extraction pits.

MNDWI — Modified NDWI

$$\text{MNDWI} = \frac{\text{Green} - \text{SWIR1}}{\text{Green} + \text{SWIR1}}$$

Better at detecting turbid (sediment-laden) water than plain NDWI. This is particularly useful in Indian rivers which carry high sediment loads.

Why it matters: Active dredging creates muddy water plumes downstream. MNDWI detects these turbidity anomalies.

BSI — Bare Soil Index

$$\text{BSI} = \frac{(\text{SWIR1} + \text{Red}) - (\text{NIR} + \text{Blue})}{(\text{SWIR1} + \text{Red}) + (\text{NIR} + \text{Blue})}$$

Detects freshly exposed, bare, dry soil or sediment. Mining sites have high BSI because:

- Topsoil and vegetation are removed
- Sand and gravel have high SWIR and Red reflectance
- No moisture-retaining vegetation cover remains

NDBI — Normalised Difference Built-up Index

$$\text{NDBI} = \frac{\text{SWIR1} - \text{NIR}}{\text{SWIR1} + \text{NIR}}$$

Originally designed to detect urban areas but also sensitive to compacted surfaces like equipment staging areas, access roads, and dried sand stockpiles.

NDTI — Normalised Difference Turbidity Index

$$\text{NDTI} = \frac{\text{SWIR1} - \text{SWIR2}}{\text{SWIR1} + \text{SWIR2}}$$

Proxy for water turbidity (sediment load). Elevated NDTI values in a river stretch indicate disturbed sediment — consistent with active dredging operations.

The Historical Feature Innovation

This is the most important technical contribution of this system.

The Core Idea

A single satellite photo cannot reliably tell us if sand mining is happening. A sandy riverbank and an active mining site can look similar in one image. But they evolve *differently over time*:

Feature	Natural Sandbar	Active Mining Site
NDVI over 3 years	Stable or seasonal	Declining (vegetation removed)
BSI over 3 years	Stable	Increasing (more bare soil)
MNDWI over 3 years	Seasonal variation	Elevated turbidity year-round
NDWI over 3 years	Follows monsoon	Anomalous during dry season

Table 2: How natural features vs. mining sites differ over time

How the Temporal Features Are Computed

For every analysis point, the system:

1. Fetches **12 quarterly Sentinel-2 images** spanning 3 years (one image per 3-month window)
2. For each image, computes the mean NDVI, BSI, MNDWI, and NDWI within a 1,500 m buffer around the point
3. Fits a **linear regression line** through the 12 values over time
4. Extracts the **slope** of that line as a feature

What the Slopes Mean

- **NDVI_trend < 0:** Vegetation declining over time → suspicious
- **BSI_trend > 0:** Bare soil increasing over time → suspicious
- **MNDWI_trend anomalous:** Water turbidity changing unexpectedly → suspicious
- **All trends stable:** Natural undisturbed location

This approach is validated by:

- Li et al. (2024) — quarterly BSI/NDWI slopes on the Mekong River achieved $F1 = 0.85$
- Bendixen et al. (2021) — temporal Landsat trends revealed 20–50% more sand extraction events than single-date analysis

Image-Level Features: What the Photo Looks Like

Beyond spectral indices, the system extracts computer vision features from the visual appearance of each satellite image chip.

Colour Statistics

For each of the Red, Green, Blue, and Grayscale channels:

- Mean brightness
- Standard deviation (contrast within the image)
- Median value

Sand mining sites tend to be brighter (high reflectance from exposed sand) and more uniform in colour than vegetated natural areas.

GLCM Texture Features

GLCM stands for **Grey Level Co-occurrence Matrix**. It analyses how pixel intensities are spatially related to their neighbours — essentially measuring the “texture” of the image.

Features extracted:

- **Contrast:** How different adjacent pixels are
- **Homogeneity:** How similar adjacent pixels are
- **Energy:** Uniformity of the texture
- **Correlation:** How linearly related adjacent pixels are
- **Dissimilarity:** Average difference between pixel pairs
- **ASM (Angular Second Moment):** Orderliness of the texture

Sand surfaces have low contrast and high homogeneity. Active mining sites often show high contrast between sand, water, and disturbed soil.

Local Binary Pattern (LBP)

LBP encodes the local texture around each pixel by comparing it to its eight neighbours. The resulting histogram captures fine-grained surface texture differences invisible to the naked eye.

Edge Detection

Sobel edge detection identifies sharp boundaries in the image. Mining sites often have:

- Sharp straight edges (cut banks, access roads)
- High edge density near equipment and extraction pits

The Machine Learning Model

What the Model Does

The model takes all the features described above — 46 features per point — and outputs a single probability between 0 and 1:

- **0.0–0.5:** No sand mining likely (shown in green on the map)
- **0.5–0.7:** Possible sand mining (shown in orange)
- **0.7–1.0:** Sand mining likely (shown in red)

Training the Model

The model learns from **examples provided by a human expert**. A domain expert looks at satellite image chips and labels each one:

- **Label 1:** “Yes, this looks like sand mining”
- **Label 0:** “No, this is a natural undisturbed area”

The system includes a dedicated **graphical labeling interface** (the GUI) that displays each image and lets the expert click a button to assign the label. Labels are saved to a CSV file for reuse.

Six Models Are Trained and Compared

To ensure the best possible model is selected, the system trains and evaluates all of the following algorithms on the same data:

Model	How It Works (Plain Language)
Random Forest	Many decision trees vote together; majority wins
Gradient Boosting	Trees built sequentially, each correcting the previous
XGBoost	Optimised gradient boosting, very fast
LightGBM	Gradient boosting optimised for large datasets
SVM	Finds the widest possible boundary between classes
Logistic Regression	Simple linear model of probability

Table 3: Machine learning models compared in the pipeline

How the Best Model Is Chosen

The system uses **5-fold cross-validation** — it splits the training data into 5 parts, trains on 4 parts and tests on 1 part, rotating through all 5 combinations. The model with the best average F1 score across all 5 folds is selected.

Why F1 Score, Not Just Accuracy?

If only 10% of locations have sand mining, a model that always says “no mining” would be 90% accurate but completely useless. F1 score penalises both missing real mining sites (false negatives) and false alarms (false positives), making it the right metric for this imbalanced problem.

The system **prefers ensemble models** (Random Forest, XGBoost, LightGBM) over SVM when scores are equal, because ensemble models generalise better to new locations they have not seen during training — particularly important with small training datasets.

Current Model Performance (Damodar Pilot)

Model	Accuracy	Precision	Recall	F1	CV F1
Random Forest	0.857	1.000	0.75	0.857	0.820
XGBoost	0.857	1.000	0.75	0.857	0.820
Gradient Boosting	0.714	1.000	0.50	0.667	0.751
LightGBM	0.571	0.571	1.00	0.727	0.714
SVM	0.714	1.000	0.50	0.667	0.820
Logistic Regression	0.857	1.000	0.75	0.857	0.880

Table 4: Model comparison results on 25 labeled training samples (Damodar pilot)

Important caveat: These results are based on only 25 labeled samples. Reliable performance estimates require 100–200+ samples. The current results should be treated as preliminary. The high CV F1 standard deviation (± 0.18 for RF) reflects this uncertainty.

The Complete Pipeline: Step by Step

Overview

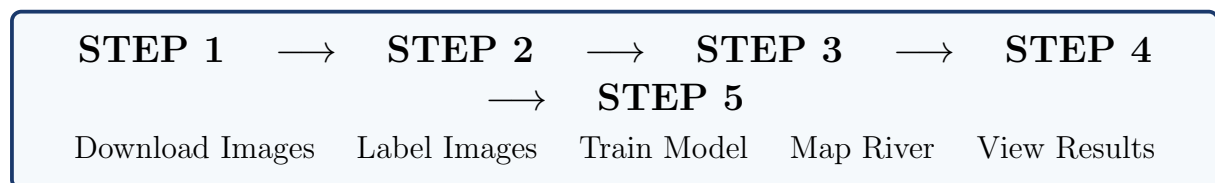


Figure 1: The five-stage pipeline from raw shapefile to interactive map

Step 1: Generating Sample Points from a Shapefile

The user provides a **river shapefile** — a GIS file that defines the path of the river as a line. The system:

1. Loads the shapefile using GeoPandas
2. Reprojects to WGS84 (standard lat/lon coordinates)
3. Walks along the river line at fixed intervals (e.g., 0.4 km)
4. Generates a list of (latitude, longitude) coordinate pairs

For the Damodar River (361 km) at 0.4 km spacing, this produces approximately **904 sample points**. For training, 30 of these are randomly selected for image download.

Step 2: Downloading Satellite Images

For each training point, the system contacts **Google Earth Engine** and:

1. Creates a 1,500 m radius circle around the point

2. Searches for the least-cloudy Sentinel-2 image within the past 12 months over that area
3. Downloads a 512×512 pixel RGB image as a PNG file
4. Applies image enhancement (contrast, sharpness, saturation) to make visual features clearer for labeling

Each image covers approximately 3×3 km on the ground and takes 4–6 seconds to download.

Step 3: Expert Labeling via the GUI

The graphical labeling interface opens automatically and displays each downloaded image one by one. The expert:

- Presses 1 for “Sand Mining” or 0 for “No Mining”
- The image automatically advances to the next
- Can press ? to skip uncertain images
- Presses ESC when done — labels are saved automatically

What to look for when labeling:

- Visible excavation pits or craters near the river
- Heavy machinery (excavators, bulldozers) on the riverbank
- Barges or boats loaded with sand in the channel
- Disturbed, irregular bare soil patches adjacent to water
- Unusually turbid or discoloured water
- Access roads cut through vegetation to the riverbank

Step 4: Feature Extraction with Historical Data

For each labeled image, the system extracts **46 features**:

Feature Group	Count	Source
Global colour statistics (mean, std, median)	12	Current image
GLCM texture features	12	Current image
LBP texture features	3	Current image
Edge detection features	3	Current image
Colour ratio features	4	Current image
Area annotation features	6	Expert highlights
Historical trend slopes	4	3–5 year GEE time series
Historical period count	1	GEE
Other derived features	1	Computed
Total	46	

Table 5: Feature breakdown used for model training and prediction

The **historical trend extraction** is the most time-consuming step — approximately 40 seconds per training point — because it requires fetching and processing 12 separate satellite images from the Earth Engine archive.

Step 5: Probability Mapping

After training, the model is applied to the full river at fine resolution (0.4–0.5 km spacing = 723–904 points). For each point:

1. A new current-date satellite image is downloaded
2. All 46 features are extracted (including 3-year history)
3. The trained model predicts the probability of sand mining
4. Results are saved to a CSV file

At 0.4 km spacing, mapping 904 points with historical analysis takes approximately **10–12 hours** (about 45 seconds per point due to the historical GEE queries).

Step 6: Output Generation

Two outputs are produced automatically:

1. **Interactive HTML Map** — an OpenStreetMap-based web map with colour-coded circles at each analysis point. Clicking any circle shows the exact probability, image date, and historical change value. The map can be opened in any web browser.
2. **Summary PNG Plot** — a four-panel figure showing: the probability distribution histogram, the classification counts bar chart, the spatial hotspot heatmap, and the top contributing features.

Resolution and Coverage

Spatial Resolution Hierarchy

Mode	Spacing	Points (Damodar)	Time	Use
Quick test	50 km	9	8 min	Debugging
Coarse	10 km	36	30 min	Initial exploration
Medium	5 km	72	1 hr	Quality check
Standard	1 km	361	4.5 hr	Regular monitoring
Fine	0.5 km	723	9 hr	Publication-quality
High-res	0.4 km	904	11 hr	Maximum detail

Table 6: Analysis resolution options and approximate runtimes for the Damodar River (361 km)

Image Chip Coverage

Each analysis point uses a **1,500 m buffer** (3 km diameter). At 0.5 km point spacing, adjacent chips overlap by 1 km. This means every location along the river is seen by at least 3 overlapping analysis windows.

Temporal Resolution

The system looks back **3–5 years** in history, sampling one image every **3 months** (quarterly). For a 3-year window this gives 12 historical data points per location. Key temporal parameters:

- **Current image:** Most recent cloud-free image (typically within 2 weeks of run date)
- **Historical window:** 3–5 years back (configurable)
- **Historical interval:** 3 months (quarterly)
- **Minimum usable periods:** 3 (system skips trend calculation if fewer are available due to cloud cover)

Scaling to All-India

Current Status

The pipeline has been designed from the start to scale to national coverage. The Damodar Basin pilot demonstrates the full technical stack. Scaling to All-India requires:

1. **More labeled training data:** Currently 25 samples. A national model needs 500–1,000 samples spanning different river types (Himalayan, Deccan, coastal, monsoon-fed)
2. **State-wise processing loop:** Process one state at a time to avoid Google Earth Engine memory limits
3. **River buffer masking:** Use HydroSHEDS or CWC river network to restrict analysis to a 2–3 km riparian corridor (reduces compute by $\sim 95\%$)
4. **Post-processing:** Convert probability points to hotspot polygons attributed to district and river name

Estimated National Scale

Parameter	Estimate
Major Indian rivers to cover	~ 400
Total river length	$\sim 200,000$ km
Points at 0.5 km spacing	$\sim 400,000$ points
Processing time (parallel)	3–5 days
GEE quota required	High Volume API
Output map resolution	0.5 km point spacing, 10 m image resolution

Table 7: Estimated parameters for All-India national mapping

Known Limitations and Future Work

Current Limitations

1. **Small training dataset (25 samples):** Reliable performance requires 100–200+ labels. The current model should be considered a proof-of-concept.
2. **No monsoon-season detection:** Sentinel-2 is blocked by clouds June–September. Sentinel-1 SAR integration is planned but not yet implemented.

3. **Only linear trends:** The historical feature uses a simple slope (linear regression). Non-linear patterns — such as episodic mining bursts or sudden onset events — are not captured. BFAST or LSTM time-series models would improve this.
4. **No independent validation:** The model has not yet been validated against independently verified ground-truth data (e.g., confirmed mining locations from news reports, court cases, or field visits).
5. **SMOTE not yet installed:** Class imbalance correction via SMOTE oversampling is coded but requires `pip install imbalanced-learn`.

Planned Upgrades

- Sentinel-1 SAR fusion for cloud-robust detection
- Active learning loop to maximise labeling efficiency
- BFAST/LandTrendr breakpoint detection for onset timing
- Foundation model features (Prithvi, SatMAE) for small dataset performance
- State-wise processing loop for All-India scaling
- Validation against NGT case locations and news archives

How to Run the System

Prerequisites

```
# Python 3.10+, Google Earth Engine account required
pip install earthengine-api geopandas pandas numpy
pip install scikit-learn xgboost lightgbm scipy
pip install Pillow requests tqdm matplotlib seaborn
pip install imbalanced-learn # for SMOTE
```

Listing 1: Installation

Authentication

```
gcloud auth application-default login
gcloud config set project YOUR_PROJECT_ID
```

Listing 2: Google Earth Engine authentication

Running the Full Pipeline

```
python scripts/run_pipeline.py \
  --mode both \
  --shapefile data/raw/your_river.shp \
  --distance 0.5 \
  --years-back 3 \
  --multiple-models \
  --sample-size 100
```

Listing 3: Full pipeline command

Key Arguments

Argument	Default	Effect
--mode	required	both / train / map / label
--shapefile	required	Path to river .shp file
--distance	0.2 km	Point spacing in km
--years-back	0	Historical window in years
--sample-size	30	Training images to download
--multiple-models	off	Compare all 6 algorithms

Table 8: Key command-line arguments

Summary

This system represents the first end-to-end, open-source pipeline for automated riverine sand mining detection in India using:

- Sentinel-2 10 m multispectral imagery
- 3–5 year quarterly historical trend features
- Six machine learning classifiers with cross-validated selection
- A human-in-the-loop labeling interface
- Interactive HTML probability maps

The Damodar Basin pilot (361 km, 25 training samples, CV $F1 = 0.82$) demonstrates technical feasibility. The architecture is explicitly designed for national scaling to All-India coverage.

The key innovation — combining *current spectral features* with *multi-year temporal trend slopes* in a single ML feature vector — addresses the core limitation of single-date sand mining detection and is grounded in recent literature from the Mekong, Ganga, and Ghana river systems.

In plain language: This system watches rivers from space for years, learns what sand mining looks like (both today and in terms of how the land has changed), and flags suspicious locations — so inspectors know exactly where to go instead of guessing.